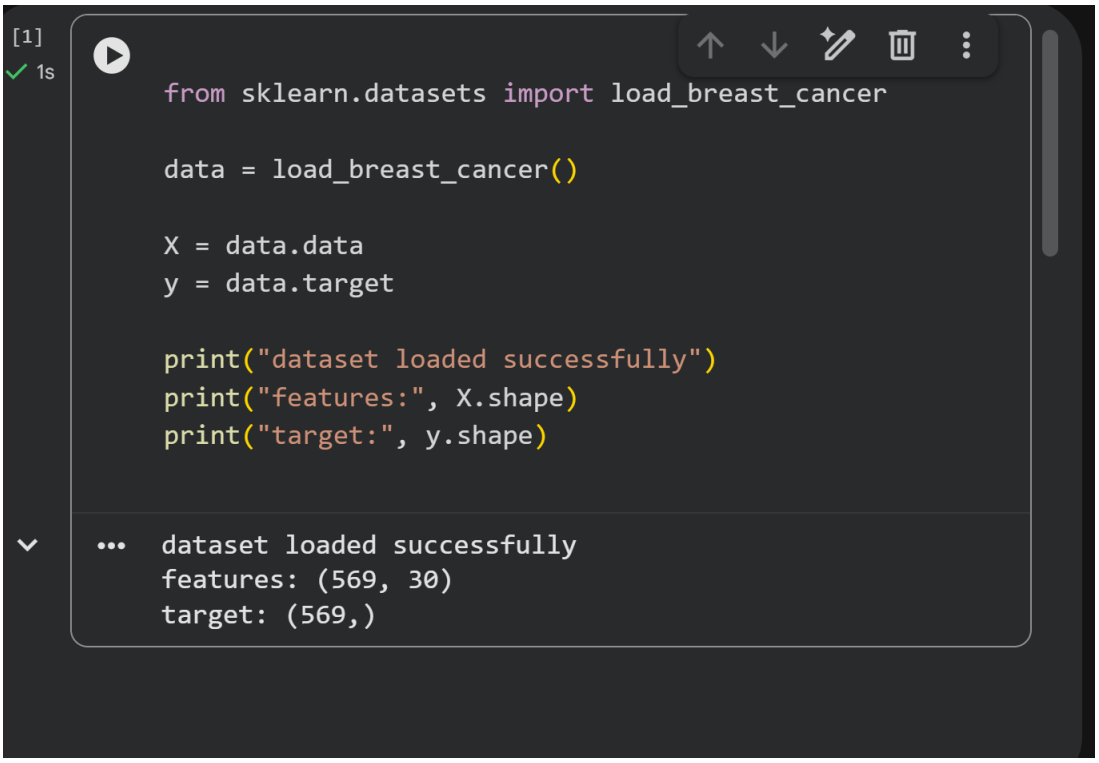
LAB # 08

KNN CLASSIFIER

Procedure:

Task 1:

Source Code And Output:



Task 2:

Source Code And Output:

```python
from sklearn.datasets import load_breast_cancer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler

data = load_breast_cancer()

X = data.data

imputer = SimpleImputer(strategy='mean')
X = imputer.fit_transform(X)

scaler = StandardScaler()
X = scaler.fit_transform(X)

print("Data preprocessing completed")
print(X[:5])
```

```
Data preprocessing completed
[[ 1.09706398e+00 -2.07333501e+00  1.26993369e+00  9.84374905e-01
   1.56846633e+00  3.28351467e+00  2.65287398e+00  2.53247522e+00
   2.21751501e+00  2.25574689e+00  2.48973393e+00 -5.65265059e-01
   2.83303087e+00  2.48757756e+00 -2.14001647e-01  1.31686157e+00
   7.24026158e-01  6.60819941e-01  1.14875667e+00  9.07083081e-01
   1.88668963e+00 -1.35929347e+00  2.30360062e+00  2.00123749e+00
   1.30768627e+00  2.61666502e+00  2.10952635e+00  2.29607613e+00
   2.75062224e+00  1.93701461e+00]
 [ 1.82982061e+00 -3.53632408e-01  1.68595471e+00  1.90870825e+00
  -8.26962447e-01 -4.87071673e-01 -2.38458552e-02  5.48144156e-01
   1.39236330e-03 -8.68652457e-01  4.99254601e-01 -8.76243603e-01
   2.63326966e-01  7.42401948e-01 -6.05350847e-01 -6.92926270e-01
  -4.40780058e-01  2.60162067e-01 -8.05450380e-01 -9.94437403e-02
   1.80592744e+00 -3.69203222e-01  1.53512599e+00  1.89048899e+00
  -3.75611957e-01 -4.30444219e-01 -1.46748968e-01  1.08708430e+00
  -2.43889668e-01  2.81189987e-01]
 [ 1.57988811e+00  4.56186952e-01  1.56650313e+00  1.55888363e+00
   9.42210440e-01  1.05292554e+00  1.36347845e+00  2.03723076e+00
   9.39684817e-01 -3.98007910e-01  1.22867595e+00 -7.80083377e-01
   8.50928301e-01  1.18133606e+00 -2.97005012e-01  8.14973504e-01
   2.13076435e-01  1.42482747e+00  2.37035535e-01  2.93559404e-01
```

Task 3:

Source Code And Output:

```python
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

data = load_breast_cancer()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("training:", X_train.shape)
print("testing:", X_test.shape)
```

```
training: (455, 30)
testing: (114, 30)
```

Task 4:

Source Code And Output:

```
[5]
✓ 0s
```

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier

data = load_breast_cancer()

X = data.data
y = data.target

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

for k in [3, 5, 7]:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)
    print("K =", k, "Model trained ")
```

```
...  K = 3 Model trained
     K = 5 Model trained
     K = 7 Model trained
```

Task 5:

Source Code And Output:

```
[6]
✓ 0s
data = load_breast_cancer()

X = data.data
y = data.target

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
```

```
Predictions: [1 0 0 1 1 0 0 0 0 1 1 0 1 1 1 0 1 1 1 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 0 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 1 1 1 0 1 0 0 0 0 0 0 1 1 1 0 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]
```

Task 6:

Source Code And Output:

```
]
0s
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

data = load_breast_cancer()

X = data.data
y = data.target

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1-Score:", f1_score(y_test, y_pred))
```

```
Accuracy: 0.9473684210526315
Precision: 0.9577464788732394
Recall: 0.9577464788732394
F1-Score: 0.9577464788732394
```

Task 7:

Source Code And Output:

```
[8]
✓ 0s
```

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

data = load_breast_cancer()

X = data.data
y = data.target

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

for k in [3, 5, 7]:
```

```python
for k in [3, 5, 7]:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    accuracy = accuracy_score(y_test, y_pred)

    print("K =", k, "Accuracy =", accuracy)
```

```
K = 3 Accuracy = 0.9473684210526315
K = 5 Accuracy = 0.9473684210526315
K = 7 Accuracy = 0.9473684210526315
```